THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 05

Full Syllabus (Chapters 1–8) | Class 9 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(c)** $(3,\ 0)$	a point on the axis is its own reflection
2	**(a)** Quadrant I	signs of (x, y) are $(+, +)$
3	**(d)** 13 units	$\sqrt{(3-(-2))^2+(-9-(3))^2}=\sqrt{25+144}=13$
4	**(d)** 6	substitute $x=0$ to get 6
5	**(a)** $4n-1$	common difference 4, so $a_n=4n-1$
6	**(b)** non-terminating repeating	denominator 15 has a prime factor other than 2 and 5
7	**(b)** rational	ratio of integers
8	**(a)** $\frac{2}{3}$	standard method ⇒ $\frac{2}{3}$
9	**(c)** $(x-9)\,(x+4)$	find a, b with $a+b$ = middle coefficient and ab = constant
10	**(b)** $(6x+1)^2$	identify $a^2+2ab+b^2=(a+b)^2$
11	**(b)** 180°	Theorem 11
12	**(b)** 105°	opposite angles of a cyclic quadrilateral add to 180°
13	**(b)** 15 cm	$d=\sqrt{17^2-8^2}=15$
14	**(c)** 22 cm	$\frac{30}{360}\times 2\pi r=22$
15	**(b)** the semi-perimeter of the triangle	$s=\frac{a+b+c}{2}$
16	**(b)** 21 cm	$r=\frac{C}{2\pi}=\frac{132\times 7}{44}=21$
17	**(b)** $\frac{5}{6}$	favourable outcomes: 1, 2, 3, 4, 6 ⇒ 5 out of 6
18	**(a)** $\frac{3}{5}$	9 favourable out of 15 equally likely balls
19	**(b)** an AP with $d=3$	constant difference 3
20	**(a)** 162	$ar^{n-1}=2\times(3)^4=162$

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	$(-2,5)$: Quadrant II; $(6,-3)$: Quadrant IV; $(0,-4)$: on the y-axis; $(-1,-1)$: Quadrant III	½ × 4
22	Slope $=-4$, y-intercept $=9$; negative slope ⇒ linear decay.	1+1
23	3 decimal places. Multiply numerator and denominator by 5^2 to get denominator 10^3.	1+1
24	$23\times 17=(20+3)(20-3)=400-9=391.$	2
25	The four points are concyclic (Theorem 10): they lie on one circle.	2
26	The square with side 10 units; area 100 sq. units (e.g. $9\times 11=99<100$).	1+1
27	$1-0.35=0.65$, since the win and not-win events together are certain (probability 1).	1+1
28	$S_{20}=\frac{20\times 21}{2}=210$; $S_{100}=\frac{100\times 101}{2}=5050$.	1+1

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	Radius $=\sqrt{1+64}=\sqrt{65}$ (check $16+49=65$, $49+16=65$ ✓). $OD^2=25+36=61<65$ ⇒ $(-5,6)$ lies **inside** the circle.	1+1+1
30	$A=13b$: 156, 130, 104 cm^2. The area changes by $13\times$ (change in b); a linear pattern with constant difference 26 for a decrease of 2 cm in b.	2+1
31	(i) rational, 9. (ii) irrational. (iii) rational, $\frac{1}{3}$. (iv) rational (repeating block 12345): $\frac{12345}{99999}=\frac{4115}{33333}$. Marking: ½+½+1+1.	½+½+1+1
32	For $a=b=c=1$: $1+1+1=3$ while RHS $=6$. So it is NOT an identity.	2+1
33	Area $=144$ cm^2 (1). Side $=\sqrt{144}=12$ cm (1). (Construction: the mean proportional of 16 and 9.)	2+1
34	(i) $\frac{2}{52}=\frac{1}{26}$ (ii) $\frac{1}{52}$ (iii) $\frac{36}{52}=\frac{9}{13}$.	1+1+1
35	$6x+12=180\Rightarrow x=28$ (2). $\angle A=108^\circ, \angle C=72^\circ$ (1).	2+1

	OR: A line meets a circle in at most two points; hence three points on a line cannot all be on one circle (equivalently, a circle can be drawn through three points only if they are non-collinear).	1+2
36	Slope between first two $= 3$; between the next two $= 6$ (2). Unequal slopes ⇒ not collinear (1).	2+1
	OR: $2 \cdot 3^{n-1} = 486 \Rightarrow 3^{n-1} = 243 = 3^5 \Rightarrow n = 6$.	2+1

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	$\frac{3}{5} = \frac{18}{30}$, $\frac{4}{5} = \frac{24}{30}$ ⇒ $\frac{19}{30}, \frac{20}{30}, \frac{21}{30}, \frac{22}{30}, \frac{23}{30}$ (3). To fit 5 integers strictly between k_1 and k_2 we need $k_2 - k_1 > 6$ — i.e. ≥ 7 (2). (With denominator 30 the difference is 6; the five values lie strictly between.)	3+2
	OR: (i) $\sqrt{3} = \frac{p}{q}$; $3q^2 = p^2$ ⇒ $3 \mid p$, $p = 3k$ ⇒ $3 \mid q$; contradiction (3). (ii) If $2 - \sqrt{3} = r$, $\sqrt{3} = 2 - r$ rational — contradiction (2).	3+2
38	Let AB be a chord with midpoint M and centre O. $OA = OB$ (radii), $AM = MB$, OM common ⇒ $\triangle OMA \cong \triangle OMB$ (SSS) (2). So $\angle OMA = \angle OMB = 90^\circ$ (1) ⇒ $OM \perp AB$ at its midpoint, i.e. O lies on the perpendicular bisector of AB (2).	2+1+2
	OR: (i) $BC = \sqrt{64 + 36} = 10$ (1). (ii) $\angle A = 90^\circ$ ⇒ BC is a diameter; centre $(4, 3)$, radius 5 (2). (iii) Distance from $(4, 3)$ to $(8, 6) = \sqrt{16 + 9} = 5$ (1). (iv) A rectangle; opposite angles are $90^\circ + 90^\circ = 180^\circ$ so it is cyclic (1).	1+2+1+1
39	Two congruent copies fit to make a parallelogram of base b and height h (2) ⇒ triangle $= \frac{1}{2}bh$ (1). Median AD: bases $BD = DC$, same height ⇒ equal areas (2).	2+1+2
	OR: (i) $s = a + b$; area $= \sqrt{(a+b)(b)(b)(a-b)} = b\sqrt{a^2 - b^2}$ (3). (ii) $h = \sqrt{25 - 9} = 4$, base 6: $\frac{1}{2} \times 6 \times 4 = 12$; formula: $3\sqrt{16} = 12$ ✓ (2).	3+2
40	(i) ₹240, ₹280, ₹320 (1). (ii) Constant difference 40; $t_n = 200 + 40n$ (2). (iii) ₹600 (1). (iv) $200 + 40n = 1000 \Rightarrow n = 20$ km (1).	1+2+1+1
	OR: (i) 1, 3, 9, 27 (1). (ii) 81, 243 (1). (iii) 3^n, a GP with $r = 3$ (1). (iv) $\frac{27}{64}$; $\left(\frac{3}{4}\right)^n$ (1). (v) It decreases towards 0, while the number of triangles increases rapidly (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTER